AI Paradoxes: What Building a Verifier — and Breaking It — Taught Me About Trusting Machines

Reflection · October 2026

This homework asked me to do two things that look opposite but turned out to be the same thing: build an AI agent that checks whether a CV is truthful, and then write a CV that fools such an agent. I built the verifier in Task 1 and it worked well — it caught every planted lie in the public test set. Then I spent about twenty minutes on Task 2 and got my own verifier to rate a CV containing four fabrications at **0.95 out of 1**. That single number is the most useful thing I have learned in this course, because it is not a story about a weak model. It is a story about how we define, measure and trust verification.

Paradox 1: a verifier is only as honest as its rulebook

My agent did not fail. It did exactly what it was told. The specification says a discrepancy lives only in name, city, jobs, education and skills — and that job descriptions, the headline and the hometown can never be the source of one. So I left every checked field perfectly truthful and put my lies where the rulebook never looks: a false hometown, a CFA charter that does not exist, an invented award, an inflated team size. The verifier reported "consistent" with high confidence. The paradox is that *faithfulness to a specification is what creates the blind spot*. When we audit an AI system by testing the cases its specification covers, we are testing the part that was already thought about; the risk lives in the definitions nobody wrote down.

Paradox 2: the attacker and the defender are the same model

Task 1 and Task 2 used the same model, the same tools, the same database and the same afternoon. My ability to build a good verifier and my ability to break one came from one and the same capability, so there is no capability threshold after which verification simply works. Both sides improve together, and what is left is a cost ratio: how many minutes of effort does one forged document cost versus one attempted detection? In recruitment and KYC, that ratio is now close to one, and it does not obviously favour the defender.

Paradox 3: delegating verification removes the witness

The reason to use an agent is that no human can read every CV. But the human was also the only place where the blind spot was visible: a person reading my attack CV would probably have asked why the hometown does not match Facebook, or what a 15-person team means for a "Manager". The agent answered with a number instead, and *0.95 reads as "this candidate is fine"*. A numeric score launders a judgement into a fact, and the more precise it looks, the less anyone asks what it measured. That is exactly how a compliance control fails silently: not with an error message, but with a confident, well-formatted score.

Paradox 4: cheaper verification produces more to verify

Making verification cheap does not reduce the amount of dishonesty; it increases the amount of screening and therefore the incentive to fabricate. Fraud was never expensive because lying is hard — it was expensive because nobody was checking at scale. My attack CV took no domain expertise at all. The predictable result is a red-queen race in which the volume of both screening and fabrication grows, and in which "we check everything" quietly becomes "we check everything we thought to define".

What I take from this

First, judge a verifier by its *coverage*, not only its accuracy: the useful question is not "how often is it right?" but "what is it structurally unable to see?". Second, always keep a human on the high-stakes decision and require the evidence, not just the score — a verifier should have to show which fields it compared and which it ignored. Third, treat a model's confidence as a claim about its own prompt, not about the world. Professionally, this has changed what I want to learn: less about building agents that produce answers, more about auditing them, writing the specifications they are judged against, and understanding where those specifications end. The uncomfortable conclusion I am left with is that my verifier was never the weak point. The rulebook was — and a rulebook with a gap in it fails with 95% confidence.

AI use: produced with the help of an AI assistant, as explicitly permitted by this task; the experiments and numbers cited are my own, from the submitted `hw2.py` run on `task2/`.